

SMART AIR QUALITY MONITORING SYSTEM USING ARDUINO UNO

Components Required

Arduino Uno
MQ-2 Gas Sensor
DHT22 Temperature & Humidity Sensor
16×2 LCD with I2C Module
Breadboard
Jumper Wires
USB Cable

Working

The Smart Air Quality Monitoring System uses an Arduino Uno with an MQ-2 gas sensor and a DHT22 temperature and humidity sensor. The MQ-2 detects smoke and combustible gases while the DHT22 measures temperature and humidity. The Arduino continuously reads both sensors and classifies the air quality as GOOD, MODERATE, or POOR based on the MQ-2 reading. The air quality status, temperature, and humidity are displayed on the 16×2 I2C LCD in real time. This project can be used in homes, classrooms, laboratories, and offices for basic indoor air monitoring.

Connections

Component	Arduino Uno Pin
MQ-2 VCC	5V
MQ-2 GND	GND
MQ-2 AO	A0
DHT22 VCC	5V
DHT22 GND	GND
DHT22 DATA	D2
LCD VCC	5V
LCD GND	GND
LCD SDA	A4
LCD SCL	A5

PROJECT LINK

<https://wokwi.com/projects/467463269627739137>